

Intelligent Handover Management for 6G LEO Satellite Constellations: A Predictive Multi-Agent PPO Approach

Abdul Samim, Attiq Ur Rehman, and KyungHi Chang* (*Senior Member, IEEE*)

Abstract—The integration of Low Earth Orbit (LEO) satellite constellations into 6G networks promises ubiquitous connectivity, yet poses unprecedented challenges for handover management due to rapid orbital motion and dynamic channel conditions. Traditional reactive handover algorithms, designed for quasi-static terrestrial networks, fail to address the multi-dimensional optimization requirements of LEO systems where satellites move at velocities exceeding 7 km/s and user-satellite connections last only 2-4 minutes. This paper proposes a Predictive Multi-Agent Proximal Policy Optimization (PMA-PPO) framework for SNR-aware load-balanced handover management in dual-layer LEO satellite networks. The framework integrates three core components: Gated Recurrent Unit (GRU) networks for temporal forecasting of channel conditions and satellite loads, distributed PPO agents for autonomous handover decision-making, and a coordination mechanism that balances signal quality with load distribution. Through comprehensive simulations of a realistic dual-layer constellation, PMA-PPO achieves significant performance improvements: up to 77.7% reduction in handover failure rates, 74.06% reduction in satellite overload duration, 35.13% improvement in throughput fairness, and ping-pong handover rates consistently below the practical 5% threshold across all load conditions, compared to state-of-the-art baseline approaches. The proposed approach achieves polynomial computational complexity versus exponential cost for exhaustive optimization, making it suitable for real-time deployment in large-scale LEO constellations.

Index Terms—6G networks, LEO satellites, handover management, multi-agent reinforcement learning, proximal policy optimization, predictive algorithms, load balancing.

I. INTRODUCTION

THE evolution toward sixth-generation (6G) wireless networks envisions integrated terrestrial-satellite architectures providing ubiquitous coverage across urban centers, remote regions, and mobile platforms [1]. LEO satellite constellations, operating at altitudes between 300–2000 km, emerge as cornerstone technology due to reduced propagation delays and global coverage capabilities [2].

Recent deployments by SpaceX Starlink (550 km altitude, 53° inclination, ~6,000 satellites), Amazon Project Kuiper (590-630 km, 33°–51.9°, 3,236 satellites), and OneWeb (1,200 km, 87.9°, 648 satellites) demonstrate the commercial feasibility of large-scale LEO mega-constellations [3]. However, these systems are optimized for *global* broadband coverage and do not specifically address the handover management and load-balancing challenges inherent to dense regional deployments over mid-latitude areas such as Northeast Asia.

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Motivated by this gap, this paper adopts a purpose-designed dual-layer research constellation distinct from any single commercial system that isolates and rigorously evaluates the handover optimization problem under realistic LEO/VLEO orbital dynamics. The constellation parameters are chosen to be consistent with 3GPP TR 38.811 NTN guidelines [6] and reflect state-of-the-art Ka-band satellite technology.

LEO handovers occur far more often than in terrestrial systems; fast 10–15 dB SNR swings plus constellation-wide load balance needs break traditional reactive schemes [5], [7]. Conventional RSSI/CHO is simple but ignores orbital predictability and induces ping-pong under dynamics [6], [9]. Optimization can be near-optimal but is intractable at scale and assumes perfect foresight [11]. RL learns policies from data, yet single-agent/reactive designs miss coordinated, predictive decisions and often optimize throughput without balancing load [8], [11], [12]. Recent work on agent-based handover frameworks in terrestrial 5G networks, such as NWDAF-orchestrated energy-aware optimization [10], further demonstrates the value of intelligent coordination mechanisms, motivating their extension to the more dynamic LEO satellite environment.

Problem Statement: At each discrete time step, every user in the dual-layer constellation must be assigned to exactly one VLEO satellite for data service, subject to per-beam capacity and minimum-SNR feasibility constraints. Because satellites move at orbital velocities exceeding 7 km/s, both the achievable SNR and each satellite's load change on the timescale of seconds, so an association decision that is optimal now may become infeasible or suboptimal within the same connection window. This creates a decision problem with two competing demands: individual users need continuous, high-quality service, while the constellation as a whole needs balanced load across satellites so that no single satellite becomes a bottleneck. A handover that improves one user's link quality can overload its target satellite and degrade service for other users on that satellite, so satellite selection cannot be made independently for each user in isolation it must account for the shared, time-varying capacity of the constellation.

This paper proposes a PMA-PPO framework that addresses these limitations by combining predictive temporal modeling with distributed multi-agent coordination. The approach leverages orbital mechanics predictability through GRU-based forecasting and distributed satellite autonomy through coordinated PPO agents, achieving polynomial complexity $\mathcal{O}(U \cdot S \cdot H \cdot D^2)$ versus exponential $\mathcal{O}(S^U)$ for exhaustive optimization, where U and S denote the numbers of users and satellites, $H = \max(H_{\text{snr}}, H_{\text{load}})$ is the prediction horizon, and D is the GRU hidden dimension.

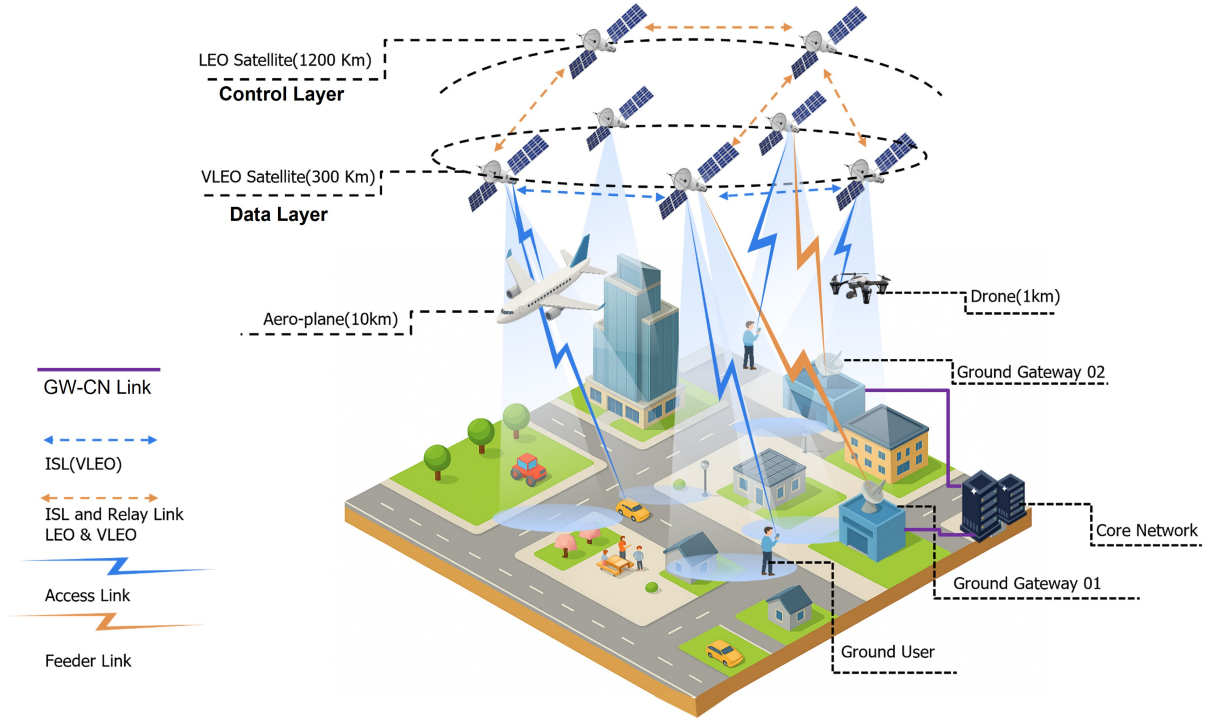


Fig. 1. Dual-layer LEO satellite architecture showing control layer and data layer with ISLs, gateways, and access to aerial and ground users.

Key Contributions:

- Novel predictive multi-agent architecture integrating GRU-based temporal forecasting with distributed PPO agents for proactive handover management
- SNR-aware load-balanced coordination mechanism with multi-objective reward optimization
- Dual-layer constellation design separating LEO control and Very Low Earth Orbit (VLEO) data functions
- Comprehensive evaluation demonstrating up to 77.7% handover failure reduction compared to CHO baseline, 74.06% load balancing improvement, and 35.13% throughput enhancement compared to state-of-the-art approaches

The remainder of this paper is organized as follows: Section II presents the dual-layer LEO network architecture and channel models. Section III formulates the multi-objective handover optimization problem. Section IV details the PMA-PPO framework architecture and algorithms. Section V provides comprehensive simulation results and performance analysis. Section VI concludes the paper and outlines future research directions.

II. SYSTEM MODEL

A. Dual-Layer LEO Constellation Architecture

We consider a dual-layer LEO satellite network architecture consisting of a control layer at higher altitude and a data layer at lower altitude, as illustrated in Fig. 1. Parameters of the system design, will be detailed in subsequent section. This hierarchical design leverages the complementary characteristics of different orbital altitudes: higher LEO satellites provide

extended coverage and longer visibility windows for control coordination, while lower VLEO satellites offer reduced propagation delay for latency-sensitive data transmission [2], [4]. While Starlink's 550 km altitude is the most widely known commercial deployment, we select 1200 km for the LEO control layer because it corresponds to the *LEO-1200* reference orbit officially defined by 3GPP TR 38.821 (Table 6.1.1.1-1) for NTN system-level simulation and channel modeling, alongside the complementary *LEO-600* reference orbit. Since our channel model, path-loss equations, and link budget (Section II-D) are calibrated against 3GPP TR 38.811/38.821, adopting the matching 3GPP reference altitude ensures internal consistency between our orbital geometry and channel model, rather than importing an altitude from a commercial constellation whose channel characteristics were not used in this study. The lower, 300 km VLEO data layer is a research-specific choice, separately justified in Section II-B by propagation-delay and spatial-reuse requirements.

For performance evaluation, we focus on a regional deployment over South Korea (approximately 600 km × 400 km coverage area) where 32 VLEO satellites from the constellation provide overlapping coverage at any given time, serving a mixed population of aerial and ground users distributed across urban and rural areas.

Realism of the User Count: With $S_{\text{act}} = 32$ active VLEO satellites, each supporting $N_b = 16$ concurrent beams (Section II-B2), the instantaneous service capacity of the visible constellation is $32 \times 16 = 512$ concurrent users. The evaluated peak load of $U = 300$ therefore corresponds to 58.6% utilization of available capacity, leaving 212 service slots free avoiding structural infeasibility and remaining consistent with

the gradual (rather than saturating) rise in handover failure rate observed in Fig. 7(a). This user count is also representative of plausible satellite-served demand for the target region: with a combined population exceeding 25 million across the Seoul Metropolitan Area, Incheon, and Busan, 300 concurrent users represents a small fraction of the region's population, consistent with NTN's role as a complementary access layer for underserved, mobile, and aerial users rather than as the primary access method for the general population. We further evaluate handover performance across a load-scaling sweep from 50 to 300 users (Fig. 7(a)) precisely to characterize behavior across this realistic range, from moderate load to near-saturation. All satellites follow circular polar orbits with inclination angles near 90° , ensuring global coverage including polar regions [13]. The orbital periods are approximately 110 minutes for LEO satellites and 90 minutes for VLEO satellites, creating a dynamic topology where satellite-user connectivity patterns evolve continuously. User terminals are distributed within a circular Region of Interest (ROI) spanning 600 km diameter as shown in practical simulation Fig. 2. Due to rapid orbital motion, user-satellite connection durations are limited, necessitating frequent handovers to maintain service continuity.

B. Layer-Specific Functionalities

The dual-layer architecture is not merely a complexity choice but a functional necessity: the VLEO data layer requires low propagation delay (≈ 1 ms) and dense spatial reuse (76 km beam footprint) that are achievable only at low altitude, while the LEO control layer requires an 4000 km coverage footprint for persistent ground-gateway visibility and continuous policy distribution, which cannot be achieved from 300 km. A single-layer constellation at 300 km would require hundreds of additional satellites to maintain persistent GW visibility, while a single-layer design at 1200 km would incur four times higher propagation delay and reduced spatial reuse. This trade-off can be quantified directly from our simulation parameters. Table III shows the dual-layer design uses 100 LEO control satellites and 300 VLEO data satellites (400 total). Critically, the control-plane cost of maintaining this second layer is small: Table VI shows the aggregate control-plane traffic across the dual-layer architecture, including all inter-layer state synchronization and policy distribution, is only 3.1 Mbps less than 0.032% of the 10 Gbps ISL capacity with round-trip control latency (10 ms) well within the 40 ms 3GPP TTT window. The dual-layer split therefore adds negligible signaling and resource overhead while avoiding the alternative costs identified above: a single 300 km layer would need substantially more satellites to match the LEO layer's persistent gateway-visibility footprint, and a single 1200 km layer would degrade data-plane latency by $4\times$ and reduce spatial reuse. Given this asymmetry large latency/coverage penalties for a single layer versus negligible control overhead for two layers the dual-layer architecture is the more resource-efficient design for this problem, not a source of added complexity. The functional separation between layers mirrors the 3GPP CU/DU split, with the LEO control layer acting as the Centralized Unit (CU) for

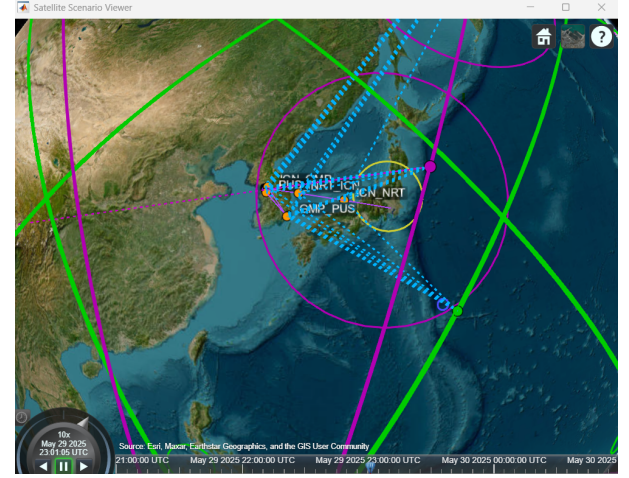


Fig. 2. ROI showing beam footprints, user service/data links, and ISLs at two instants: time t .

policy intelligence and the VLEO data layer as the Distributed Unit (DU) for user-plane execution.

1) *LEO Control Layer*: The LEO control layer operates as a distributed coordination platform, hosting the predictive algorithms and multi-agent decision-making processes described in Section IV. Each LEO satellite maintains global network state information including:

- Real-time user-satellite SNR measurements and predictions
- VLEO satellite load distributions and capacity headroom forecasts
- Handover history and mobility pattern analysis
- Inter-layer coordination messages and policy updates

The extended coverage footprint of LEO satellites (approximately 2000 km diameter at minimum elevation angle of 10°) enables persistent visibility with ground infrastructure, facilitating continuous state synchronization and policy distribution [5].

2) *VLEO Data Layer*: The VLEO data layer provides high-throughput user access through spot beam technology. Each VLEO satellite is equipped with $N_b = 16$ electronically steerable spot beams, each allocated 20 MHz bandwidth in Ka-band frequencies [14]. The beam footprint diameter is approximately 76 km on Earth's surface, corresponding to a half-power beamwidth of 7° - 8° . The tessellation of 16 beams ensures complete coverage of the 600 km ROI when the satellite is overhead.

We adopt Frequency Division Duplex (FDD) operation with a 20 GHz downlink and 27.5 GHz uplink carrier, consistent with Ka-band NTN specifications in 3GPP TR 38.821. A frequency reuse factor of 1 is applied across the $N_b = 16$ spot beams. Each beam has a half-power beamwidth of 7° - 8° , corresponding to a footprint diameter of 76 km at 300 km altitude. Adjacent beam centers are separated by 76 km on the ground, yielding a spatial isolation exceeding 25 dB for Ka-band antenna patterns [13], which renders inter-beam interference negligible relative to the dominant thermal noise floor. This is consistent with the single-layer interference

TABLE I
LIST OF MAIN MATHEMATICAL NOTATIONS

Symbol	Description	Symbol	Description
<i>Sets and Indices</i>			
$\mathcal{U}, S$	User & satellite sets	U, S	# users & satellites
$t \in \{1, \dots, T\}$	Discrete time index	$\mathcal{U}_s(t)$	Users served by sat. s
<i>Association and Feasibility</i>			
$x_{u,s}(t)$	Association decision	$\mathbf{x}(t)$	Association vector
$\chi_{u,s}(t)$	SNR feasibility flag	$\mathcal{C}_{u,s}(t)$	Candidate set
$\mathbf{s}_s^{(t)}$	Agent s state obs.	$\mathbf{I}_{\text{candidate}}^{(t)}$	Candidate avail. indicator
<i>Channel and Link Budget</i>			
$\text{SNR}_{u,s}(t)$	Instantaneous SNR (dB)	$\widehat{\text{SNR}}_{u,s}(t+h)$	Predicted SNR
$\rho_s(t)$	Satellite s load factor	$\widehat{\rho}_s(t+h)$	Predicted load
$\bar{\rho}(t)$	Network avg. load	$d_{u,s}(t)$	User-sat. distance (km)
$\dot{d}_{u,s}(t)$	Rate of distance change	$\theta_{u,s}(t)$	Elevation angle
$\tau_{u,s}^{\text{rem}}(t)$	Remaining visibility (s)	$\Theta_{\min}$	Min SNR threshold (dB)
P_{tx}	Transmit power (dBW)	$G_{\text{sat}}, G_{\text{user}}$	Antenna gains (dBi)
$L_{\text{fs}}, L_{\text{sh}}, L_{\text{atm}}$	Path, shadow, atm. loss (dB)	N_0, W	Noise density, bandwidth
N_b	Beams per satellite	σ_{sh}	Shadowing std. dev. (dB)
<i>GRU Predictor</i>			
$\mathbf{h}^{(t)}$	GRU hidden state	$H_{\text{snr}}, H_{\text{load}}$	Pred. horizons (steps)
$H = \max(H_{\text{snr}}, H_{\text{load}})$	Unified horizon	D	GRU hidden dimension
w	Historical window length	d_x	GRU input dimension
$\mathbf{r}^{(t)}, \mathbf{z}^{(t)}$	Reset & update gates	$\tilde{\mathbf{h}}^{(t)}$	GRU candidate state
$\mathbf{N}_{\text{HO},s}(t)$	HO arrival history	$\mathbf{v}_{\text{user}}(t)$	User mobility vectors
<i>PPO and Learning</i>			
$\pi_\theta(\mathbf{a} \mathbf{s})$	Policy network	$V_\phi(\mathbf{s})$	Value network
L	Network depth (layers)	N	Network width (neurons)
$\hat{A}_t$	Advantage estimate	$r_t(\theta)$	Probability ratio
γ	Discount factor	ϵ	PPO clip parameter
α_π, α_v	Actor & critic LR	λ	GAE parameter
N_{hyp}	Hypernet network dim. (QMIX)		
<i>Reward and Objectives</i>			
$R_s(t)$	Total reward (sat. s)	w_1, w_2, w_3	Reward weights
$\mathcal{J}_1, \mathcal{J}_2, \mathcal{J}_3$	Throughput, HO, load obj.	α, β	Load & stability coefficients
$H_u(t)$	Aggregate HO indicator	$H_{u,s}(t)$	Per-sat. HO indicator
$\tau_{\min}$	Min. dwell threshold (s)	$\text{SNR}_{\max}$	Max SNR reference (dB)

$\mathbb{I}[\cdot]$: indicator function; $\odot$: element-wise product; $\sigma(\cdot)$: sigmoid.

model adopted in 3GPP TR 38.811 [6]. Each beam serves at most one user at a time under Time-Division Multiple Access (TDMA) scheduling within a beam, yielding a maximum capacity of $N_b = 16$ concurrent users per VLEO satellite. This constraint cleanly decouples intra-beam scheduling from the inter-satellite handover optimization problem that is the

primary focus of this work; extension to multi-user beams with NOMA or MU-MIMO is left as future work.

3) *Inter-Satellite Link (ISL) Architecture*: The constellation employs optical free-space laser ISLs operating at $\lambda = 1550$ nm with a per-link capacity of 10 Gbps and a maximum range of 5000 km, consistent with state-of-the-art laser ISL (LISL) implementations [23]. At VLEO/LEO altitudes above 100 km, atmospheric absorption of the optical beam is negligible, and pointing-acquisition-tracking (PAT) systems maintain link availability exceeding 99.9% under nominal orbital conditions [17]. Accordingly, ISL propagation delay is modeled as $\tau_{\text{ISL}} = d_{\text{ISL}}/c$, where d_{ISL} is the inter-satellite distance and c is the speed of light. The inter-layer ISL distance d_{ISL} is determined by the relative geometry of the LEO and VLEO shells and is bounded as

$$d_{\text{ISL}}^{\min} = h_{\text{LEO}} - h_{\text{VLEO}},$$

$$d_{\text{ISL}}^{\max} = \sqrt{r_{\text{LEO}}^2 + r_{\text{VLEO}}^2 - 2r_{\text{LEO}}r_{\text{VLEO}}\cos\theta_c}. \quad (1)$$

Here θ_c denotes the maximum geocentric (central) angle between the LEO and VLEO sub-satellite points for which the inter-layer line of sight is not obstructed by the Earth i.e., the grazing condition at which the line connecting the two satellites is tangent to Earth's surface. This angle is given in closed form by $\theta_c = \arccos(R_\oplus/r_{\text{VLEO}}) + \arccos(R_\oplus/r_{\text{LEO}})$. With $h_{\text{LEO}} = 1200$ km, $h_{\text{VLEO}} = 300$ km, and $R_\oplus = 6371$ km, this gives $\theta_c \approx 49.9^\circ$, yielding a minimum inter-layer ISL length of $d_{\text{ISL}}^{\min} = 900$ km (satellites directly aligned radially) and a geometric maximum of $d_{\text{ISL}}^{\max} \approx 6067$ km at θ_c . Since this geometric maximum exceeds the 5000 km hardware range of the laser ISL terminals specified above, the achievable inter-layer link range is in practice bounded by hardware capability rather than Earth-blockage geometry: the 5000 km hardware limit corresponds to a central angle of only $\theta_c \approx 40.5^\circ$, well inside the 49.9° geometric horizon limit. Accordingly, inter-layer ISLs are established only between LEO-VLEO pairs within this hardware-limited range. where $r_{\text{LEO}} = R_\oplus + h_{\text{LEO}}$ and $r_{\text{VLEO}} = R_\oplus + h_{\text{VLEO}}$. Hence, the one-way ISL propagation delay is

$$\tau_{\text{ISL}} = \frac{d_{\text{ISL}}}{c} \quad (2)$$

The round-trip control latency, including access delay, ISL delay, and onboard processing, is given by

$$\tau_{\text{RT}} = \tau_{\text{access}} + 2\tau_{\text{ISL}} + \tau_{\text{proc}}$$

$$\leq \frac{h_{\text{VLEO}}}{c} + \frac{2d_{\text{ISL}}^{\max}}{c} + \tau_{\text{proc}} \quad (3)$$

where $\tau_{\text{access}} = h_{\text{VLEO}}/c$. Control message overhead (e.g., state synchronization and policy updates) consumes a negligible fraction ($<1\%$) of the 10 Gbps ISL capacity, so ISL throughput does not constitute a bottleneck for handover signaling. Each satellite maintains five ISL connections:

- **Intra-plane ISLs**: Bidirectional links to adjacent satellites within the same orbital plane
- **Inter-plane ISLs**: Cross-links to satellites in neighboring orbital planes

These assumptions follow the established modeling convention for constellation-level handover studies [22], allowing focus

on the handover management problem without ISL protocol complexities, while remaining realistic for modern laser inter-satellite communication systems [23].

C. User Traffic and Mobility Model

The system serves mobile users uniformly distributed within the ROI. Users follow random waypoint mobility with velocities up to 1000 km/h, representing mixed pedestrian, vehicular, and aerial platform scenarios [18].

We employ a full-buffer traffic model where each user maintains continuous data transmission/reception requests [6]. This saturation traffic assumption represents worst-case loading conditions for evaluating handover algorithm performance under stress. Quality-of-Service requirements focus on:

- Service continuity: Minimizing handover interruption time
- Throughput fairness: Balancing capacity allocation across users
- Coverage maintenance: Ensuring minimum SNR thresholds

D. Service and Application Models

The full-buffer traffic model adopted in this work is intended to represent enhanced Mobile Broadband (eMBB) service e.g., high-throughput data, video streaming, and general-purpose internet access which is the primary target service class for Ka-band NTN access as defined in 3GPP TR 38.821. Ground users represent fixed and pedestrian/vehicular broadband subscribers across the Seoul Metropolitan Area, Incheon, and Busan, while aerial users represent broadband connectivity for airborne platforms along commercial flight corridors (e.g., passenger in-flight connectivity), consistent with the aerial-user scenarios described in 3GPP TR 38.811. We adopt full-buffer traffic as a deliberate worst-case proxy for eMBB demand: because our goal is to stress-test handover reliability and load-balancing performance under sustained high load (Section II-C), rather than to model service-specific traffic bursts, a saturating full-buffer source provides an upper bound on handover and overload stress relative to any bursty or intermittent eMBB traffic pattern. Latency-sensitive or low-rate services (e.g., URLLC, mMTC/IoT) are outside the scope of this study and are noted as future work in Section VI.

E. Channel Model and Link Budget

1) *Propagation Environment*: User-satellite communication employs Ka-band downlink carriers at 20 GHz with 20 MHz channel bandwidth per beam [16]. The Ka-band frequency at 20 GHz provides high data throughput capabilities while maintaining reasonable atmospheric attenuation for LEO satellite communications [19].

2) *Path Loss Modeling*: The received signal power follows the standard satellite link equation [19]:

$$P_{u,s}(t) = P_{tx} + G_{sat} + G_{user} - L_{fs}(t) - L_{sh}(t) - L_{atm} \quad (4)$$

where the free-space path loss is:

$$L_{fs}(t) = 20 \log_{10} \left(\frac{4\pi d_{u,s}(t)f}{c} \right) [\text{dB}] \quad (5)$$

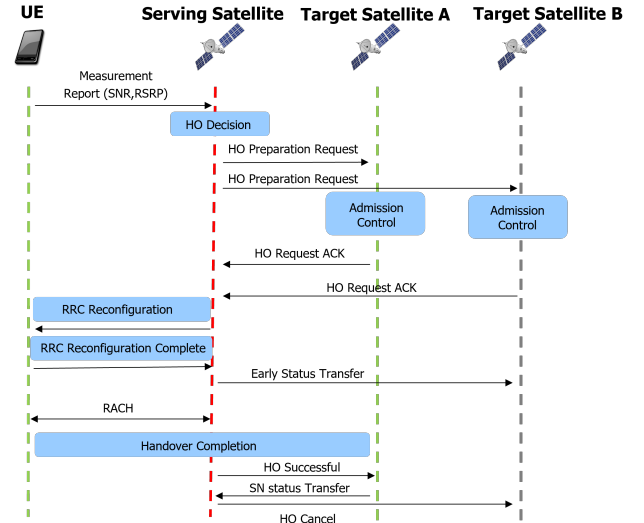


Fig. 3. Conditional handover call flow as defined in 3GPP.

with $f = 20 \times 10^9$ Hz as the Ka-band carrier frequency and $c = 3 \times 10^8$ m/s as the speed of light.

Shadowing effects are modeled as log-normal fading with zero mean and standard deviation $\sigma_{sh} = 4$ dB, representing atmospheric turbulence and small-scale obstacles [15]:

$$L_{sh}(t) \sim \mathcal{N}(0, \sigma_{sh}^2) \quad (6)$$

Atmospheric absorption $L_{atm} \approx 0.7$ dB accounts for oxygen and water vapor attenuation at 20 GHz Ka-band frequencies [19]. Ka-band systems experience moderate atmospheric losses while providing significantly higher bandwidth compared to lower frequency bands.

3) *SNR and Capacity Calculations*: The instantaneous downlink SNR is computed as:

$$\text{SNR}_{u,s}(t) = \frac{P_{u,s}(t)}{N_0 W} = \frac{P_{tx} G_{sat} G_{user}}{L_{total}(t) \cdot N_0 W} \quad (7)$$

where $L_{total}(t) = 10^{(L_{fs}(t) + L_{sh}(t) + L_{atm})/10}$ represents the combined path loss in linear scale, and all antenna gains are converted from dB to linear scale.

Service availability requires $\text{SNR}_{u,s}(t) \geq \Theta_{min} = 8$ dB, corresponding to the minimum threshold for reliable QPSK demodulation with forward error correction [20]. The achievable data rate follows Shannon's capacity formula:

$$C_{u,s}(t) = W \log_2(1 + \text{SNR}_{u,s}(t)) [\text{bps}] \quad (8)$$

This upper bound guides optimization algorithms while practical systems employ adaptive modulation and coding (AMC) to approach capacity limits [6].

The 3GPP Conditional Handover (CHO) call flow illustrated in Fig. 3 follows the standardized procedure defined in 3GPP TS 38.331 [21]. As shown, the UE first transmits a Measurement Report containing SNR/RSRP readings from both the serving and neighboring satellites. Upon satisfying a configured threshold condition (event A3 or A5), the serving satellite issues parallel Handover Preparation Requests to one or more candidate target satellites. Each candidate

TABLE II
TECHNICAL RATIONALE: PMA-PPO VS. REPRESENTATIVE LEO HANDOVER FAMILIES

Approach	Timing	Key Inputs (State)	Coordination	Spec-A: Forecasting	State	Spec-B: HO Decision Style	Primary LEO/VLEO	Limitation	in Ref.
3GPP CHO / RSSI	Reactive	Instant RSRP/SNR thresholds, hysteresis	None (per-UE)	None (only)	(snapshot)	Threshold trigger (e.g., A3/A5)	No foresight; ping-pong and overload under fast topology changes and moving beams		[6]
Optimization (MILP/MINLP)	Predictive (assumed)	Forecast SNR/capacity, link constraints, load caps	Centralized solver	Full-horizon (idealized)	oracle	Global assignment (batch)	Intractable at constellation scale; brittle to model mismatch/forecast errors		[31]
Single-Agent RL	Reactive	Local history of SNR/RSRQ, HO events	None	None (learns from history)		Per-UE stay/HO action	Myopic policies; non-stationarity causes instability and congestion spirals		[32]
Multi-Agent RL (reactive)	Reactive	Local SNR/load snapshots per sat./cell	Weak (post-hoc rules)	Short only	snapshots	Decentralized per-agent action	Conflict/oscillation without foresight; ad-hoc conflict resolution		[33]
PMA-PPO (Proposed)	Predictive H-step	GRU-based SNR/load forecasts, dwell/feasibility filters	SNR-aware load-balanced coordinator	Short-horizon GRU forecasts (H steps)		Constrained policy over candidate set	Depends on predictor quality; This designed for real-time with work polynomial scaling		

performs Admission Control and responds with a Handover Request ACK. The UE then undergoes RRC Reconfiguration, accesses the selected target satellite via RACH, and the handover is finalized through a status transfer or cancelled (*HO Cancel*) if link conditions recover before execution. This reactive, threshold-based baseline exposes the core limitation motivating PMA-PPO: it cannot anticipate rapid SNR drops or satellite overload in LEO environments, and the parallel preparation across multiple candidates introduces signaling overhead that degrades performance under the fast topology changes characteristic of VLEO/LEO constellations.

To focus on handover optimization challenges, we assume:

- 1) **Perfect State Information:** All network elements have access to current system state.
- 2) **Synchronized Operation:** Control decisions are made with globally consistent timing across the constellation
- 3) **Reliable Control Channels:** ISL networks provide error-free delivery of handover commands and status updates

These idealizations are reasonable for modern satellite systems with precise orbit determination, GPS timing synchronization, and robust laser ISL implementations [17], [23].

III. PROBLEM FORMULATION

The dual-layer LEO satellite architecture described in Section II presents a handover management challenge that fundamentally differs from terrestrial cellular systems. While terrestrial networks assume quasi-static base station positions, LEO satellites exhibit orbital velocities exceeding 7 km/s, creating rapidly evolving channel conditions and load distributions that invalidate conventional reactive handover algorithms [2]. The problem is further complicated by the dual-layer coordination between control decisions at the LEO layer and data transmission at the VLEO layer.

Concretely, this means each satellite association decision must rely on *predicted* future channel and load conditions rather than the current instantaneous state, since by the time a reactive decision is executed the underlying conditions may already have changed. In addition to the service-quality and load-balancing requirements above, a good policy must also

avoid switching users between satellites more often than necessary, since each handover interrupts service. We now formalize this as a constrained sequential decision problem. Consider the system at discrete time instances $t \in \{1, 2, \dots, T\}$ with user set $\mathcal{U} = \{1, 2, \dots, U\}$ and VLEO satellite set $\mathcal{S} = \{1, 2, \dots, S\}$ for data transmission. The handover management seeks optimal association decisions $\mathbf{x}(t) = \{x_{u,s}(t)\}$ where:

$$x_{u,s}(t) = \begin{cases} 1, & \text{if user } u \text{ associates with satellite } s \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

subject to fundamental constraints:

$$\sum_{s \in \mathcal{S}} x_{u,s}(t) \leq 1, \quad \forall u \in \mathcal{U} \quad (10)$$

$$\sum_{u \in \mathcal{U}} x_{u,s}(t) \leq N_b, \quad \forall s \in \mathcal{S} \quad (11)$$

$$x_{u,s}(t) \leq \chi_{u,s}(t), \quad \chi_{u,s}(t) = \mathbb{I}[\text{SNR}_{u,s}(t) \geq \Theta_{\min}] \quad (12)$$

Constraint (10) enforces single satellite association per user at each time step, preventing split-beam transmission. Constraint (11) limits each satellite to at most $N_b = 16$ simultaneous users, reflecting the physical beam capacity of the VLEO spot beam array. Constraint (12) enforces a minimum signal quality requirement: user u may only associate with satellite s if the instantaneous SNR exceeds the demodulation threshold $\Theta_{\min} = 8$ dB, where $\mathbb{I}[\cdot]$ is the indicator function. Together, these three constraints define the feasible association set at each time step, which forms the action space over which the PMA-PPO policy is optimized.

The system state encompasses predicted channel quality, load distributions, and geometric relationships:

$$\mathbf{s}(t) = \{\widehat{\text{SNR}}_{u,s}(t+1), \hat{\rho}_s(t+1), \tau_{u,s}(t)\} \quad (13)$$

where $\hat{\rho}_s(t) = \frac{1}{N_b} \sum_{u \in \mathcal{U}} x_{u,s}(t)$ represents predicted satellite load factor.

The handover management involves simultaneous optimization of three competing objectives [24]:

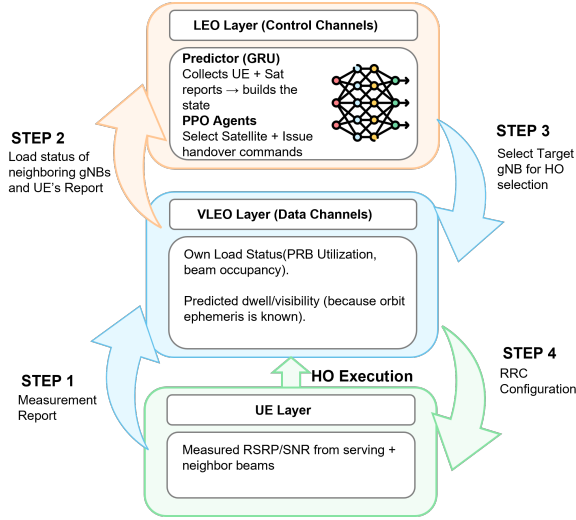


Fig. 4. LEO-VLEO handover control stack.

Network Throughput Maximization:

$$\mathcal{J}_1 = \mathbb{E} \left[\sum_{t=1}^T \sum_{u \in \mathcal{U}, s \in \mathcal{S}} x_{u,s}(t) W \log_2(1 + \text{SNR}_{u,s}(t)) \right] \quad (14)$$

Handover Frequency Minimization:

$$\mathcal{J}_2 = \mathbb{E} \left[\sum_{t=2}^T \sum_{u \in \mathcal{U}} H_u(t) \right], \quad (15)$$

$$H_u(t) = \mathbb{I} \left[\sum_{s \neq s'} x_{u,s}(t-1) x_{u,s'}(t) \geq 1 \right].$$

Load Balancing Across Satellites:

$$\mathcal{J}_3 = \mathbb{E} \left[\sum_{t=1}^T \left(\max_{s \in \mathcal{S}} \rho_s(t) - \min_{s \in \mathcal{S}} \rho_s(t) \right)^2 \right] \quad (16)$$

The composite optimization problem becomes:

$$\max_{\pi} w_1 \mathcal{J}_1 - w_2 \mathcal{J}_2 - w_3 \mathcal{J}_3 \quad (17)$$

Rapid Channel Dynamics: LEO satellite motion creates SNR variations of 10-15 dB within 2-4 minute connection windows. The SNR evolution exhibits both deterministic geometric components and stochastic fading [26]:

$$\text{SNR}_{u,s}(t+1) = f_{geo}(d_{u,s}(t+1), \theta_{u,s}(t+1)) + h_{fading}(t+1) \quad (18)$$

Scalability Constraints: For U users and S satellites, exhaustive optimization exhibits $O(S^U)$ complexity, making real-time solutions intractable for practical system dimensions.

Prediction Requirements: Optimal handover timing requires accurate forecasting of future channel conditions and satellite loads, necessitating predictive algorithms that can handle temporal dependencies in satellite communications.

IV. PROPOSED PMA-PPO FRAMEWORK

A. Training Data Source

Both the GRU predictors and the PPO agents are trained entirely on synthetic trajectories generated by the same MATLAB Satellite Communications Toolbox simulation environment used for performance evaluation (Section V-A), which implements the 3GPP TR 38.811 orbital geometry and channel model. No external or real-world satellite telemetry is used. Training trajectories are produced by running the dual-layer constellation simulator forward in time under the user mobility and full-buffer traffic models described in Section II-C, recording the resulting SNR, load, and handover event sequences as supervised targets for the GRU predictors and as environment transitions (s_t, a_t, r_t, s_{t+1}) for PPO (Algorithm 1). To prevent data leakage, training and evaluation use independent simulation episodes generated from different random seeds governing user initial positions, mobility trajectories, and shadowing realizations; results reported in Section V are computed exclusively on held-out evaluation episodes not seen during training.

The technical challenges identified in Section III—rapid channel dynamics, exponential computational complexity $O(S^U)$, and coordination requirements across distributed satellites—expose fundamental limitations of existing handover approaches. Table II provides a comprehensive technical analysis comparing representative handover families, revealing that conventional approaches suffer from critical deficiencies: reactive timing mechanisms that cannot anticipate orbital dynamics, lack of constellation-wide load balancing, and inadequate multi-agent coordination for distributed satellite decision-making.

Figure 4 illustrates the dual-layer control architecture, where the LEO control layer hosts predictive algorithms and coordination logic, while the VLEO data layer executes handover decisions through distributed PPO agents. To address the above discussed limitations, we propose a Predictive Multi-Agent Proximal Policy Optimization (PMA-PPO) framework illustrated in Figure 5 that integrates three synergistic components: (1) GRU-based temporal forecasting for SNR and load prediction with H -step lookahead, (2) distributed PPO agents enabling autonomous satellite decision-making with polynomial complexity $\mathcal{O}(U \cdot S \cdot H \cdot D^2)$, and (3) SNR-aware load-balanced coordination mechanisms for conflict-free handover execution.

B. Predictive Module Architecture

1) *Rationale for GRU-based Forecasting:* Traditional time-series forecasting methods like ARIMA or Kalman filters assume linear dynamics and Gaussian noise, which inadequately model the complex nonlinear relationships between satellite motion, channel propagation, and user mobility in LEO systems [27]. GRUs are selected for their superior capability to:

- Capture long-term temporal dependencies in satellite pass dynamics
- Handle nonlinear channel evolution due to atmospheric effects

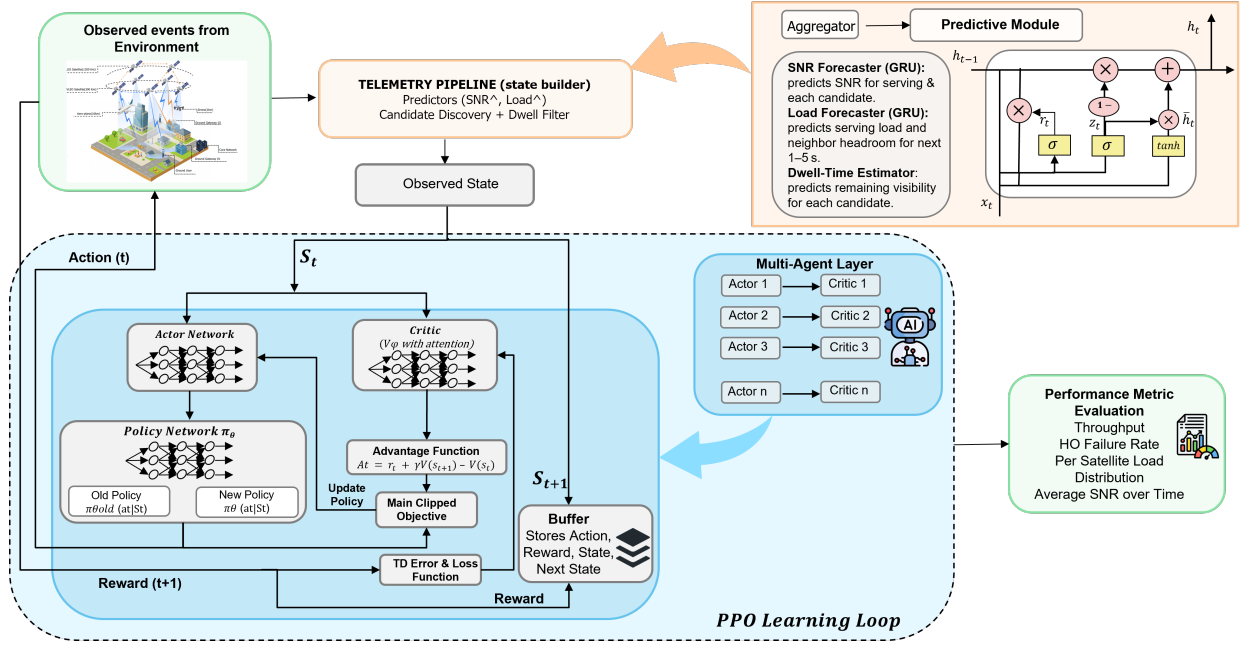


Fig. 5. Telemetry-driven Predictive Multi-agent PPO handover architecture.

- Adapt to different orbital geometries through learned representations
- Provide computational efficiency compared to LSTM networks [28]

2) *SNR Forecasting via GRU Networks*: The SNR forecaster processes historical observations to predict future channel quality. The GRU architecture implements the following update equations [27]:

$$\mathbf{r}^{(t)} = \sigma(\mathbf{W}_r \mathbf{x}^{(t)} + \mathbf{U}_r \mathbf{h}^{(t-1)} + \mathbf{b}_r) \quad (19)$$

$$\mathbf{z}^{(t)} = \sigma(\mathbf{W}_z \mathbf{x}^{(t)} + \mathbf{U}_z \mathbf{h}^{(t-1)} + \mathbf{b}_z) \quad (20)$$

$$\tilde{\mathbf{h}}^{(t)} = \tanh(\mathbf{W}_h \mathbf{x}^{(t)} + \mathbf{U}_h (\mathbf{r}^{(t)} \odot \mathbf{h}^{(t-1)}) + \mathbf{b}_h) \quad (21)$$

$$\mathbf{h}^{(t)} = (1 - \mathbf{z}^{(t)}) \odot \mathbf{h}^{(t-1)} + \mathbf{z}^{(t)} \odot \tilde{\mathbf{h}}^{(t)} \quad (22)$$

Input features combine SNR history with geometric indicators:

$$\mathbf{x}^{(t)} = [\text{SNR}_{u,s}(t-w:t), d_{u,s}(t-w:t), \dot{d}_{u,s}(t), \cos(\theta_{u,s}(t))]^T \quad (23)$$

where w is the historical window length (number of past time steps retained in the input sequence), $\dot{d}_{u,s}(t)$ is the rate of change of the user-satellite distance, $\theta_{u,s}(t)$ is the elevation angle, and $d_x = 4$ is the GRU input feature dimension. The GRU gates $\mathbf{r}^{(t)}$ and $\mathbf{z}^{(t)}$ in (19)–(20) are the reset and update gates, respectively, and $\tilde{\mathbf{h}}^{(t)}$ in (21) is the candidate hidden state.

Multi-step SNR predictions are generated through:

$$\widehat{\text{SNR}}_{u,s}(t+1:t+H_{\text{snr}}) = \text{MLP}_{\text{SNR}}(\mathbf{h}^{(t)}) \quad (24)$$

3) *Load Forecasting*: Satellite load evolution depends on user mobility patterns and handover decisions. The load

predictor processes utilization history and handover arrival patterns:

$$\hat{\rho}_s(t+1:t+H_{\text{load}}) = \text{GRU}_{\text{load}}([\rho_s(t-w:t), \mathbf{N}_{\text{HO},s}(t-w:t), \mathbf{v}_{\text{user}}(t)]) \quad (25)$$

where $\mathbf{N}_{\text{HO},s}(t)$ represents handover arrival history and $\mathbf{v}_{\text{user}}(t)$ captures user mobility vectors.

4) *Dwell Time Estimation*: Leveraging orbital mechanics, remaining visibility duration is computed analytically [29]:

$$\tau_{u,s}^{\text{rem}}(t) = \frac{\arccos(\sin \phi_u \sin \phi_s + \cos \phi_u \cos \phi_s \cos \Delta\lambda) - \theta_{\min}}{|\omega_s|} \quad (26)$$

where ϕ_u and ϕ_s are user and satellite latitudes, $\Delta\lambda$ is the longitude difference, $\theta_{\min}$ is the minimum elevation angle, and ω_s is the satellite angular velocity.

C. Multi-Agent PPO Framework

Each VLEO satellite operates an independent PPO agent with shared environmental feedback for coordination [25].

1) *Agent State Representation*: Agent s observes predicted system conditions:

$$\mathbf{s}_s^{(t)} = [\hat{\text{SNR}}_s^{(t+1:t+H)}, \hat{\rho}_{-s}^{(t+1)}, \tau_s^{(t)}, \mathbf{I}_{\text{candidate}}^{(t)}] \quad (27)$$

where $\hat{\rho}_{-s}^{(t)}$ represents predicted neighbor loads and $\mathbf{I}_{\text{candidate}}^{(t)}$ indicates candidate availability after dwell filtering.

2) *Policy and Value Networks*: The policy network parameterizes handover action probabilities using softmax activation [24]:

$$\pi_\theta(\mathbf{a}_s | \mathbf{s}_s) = \text{softmax}(\mathbf{W}_\pi \phi(\mathbf{s}_s) + \mathbf{b}_\pi) \quad (28)$$

Algorithm 1 PPO Agent Training Process

Require: Policy network π_θ , value network V_ϕ , experience buffer $\mathcal{B}$

Ensure: Updated parameters θ', ϕ'

- 1: **Experience Collection:**
- 2: **for** each episode **do**
- 3: **for** each time step t **do**
- 4: $a_t \sim \pi_\theta(s_t)$ {Sample action}
- 5: Execute a_t , observe r_t, s_{t+1}
- 6: Store (s_t, a_t, r_t, s_{t+1}) in $\mathcal{B}$
- 7: **end for**
- 8: **end for**
- 9: **Advantage Computation:**
- 10: **for** each transition (s_t, a_t, r_t, s_{t+1}) in $\mathcal{B}$ **do**
- 11: $\delta_t \leftarrow r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$
- 12: $A_t \leftarrow \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}$ {GAE}
- 13: **end for**
- 14: **PPO Update:**
- 15: **for** $k = 1$ to K epochs **do**
- 16: **for** each mini-batch in $\mathcal{B}$ **do**
- 17: $r_t(\theta) \leftarrow \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$
- 18: $L^{CLIP} \leftarrow \mathbb{E}[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)]$
- 19: $L^{VF} \leftarrow \mathbb{E}[(V_\phi(s_t) - \hat{V}_t)^2]$
- 20: $\theta \leftarrow \theta + \alpha_\pi \nabla_\theta L^{CLIP}$
- 21: $\phi \leftarrow \phi - \alpha_v \nabla_\phi L^{VF}$
- 22: **end for**
- 23: **end for**
- 24: **return** θ', ϕ'

The value function estimates expected return:

$$V_\phi(\mathbf{s}_s) = \mathbf{W}_v \phi(\mathbf{s}_s) + b_v \quad (29)$$

where $\phi(\cdot)$ represents the feature extraction function and $\mathbf{s}_s$ is the state observation for satellite s .

3) *PPO Loss Function:* The clipped surrogate objective prevents destructive policy updates [25]:

$$L^{CLIP}(\theta) = \mathbb{E} \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (30)$$

where $r_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)}{\pi_{\theta_{old}}(\mathbf{a}_t|\mathbf{s}_t)}$ and $\hat{A}_t$ is the generalized advantage estimate [30].

D. SNR-Aware Load-Balanced Coordination

The coordination mechanism integrates predicted SNR conditions with load balancing objectives through a composite reward function that guides distributed agents toward globally optimal solutions.

1) *Multi-Objective Reward Design:* The reward function balances throughput, handover cost, and load fairness:

$$R_s(t) = w_1 R_{snr}(t) + w_2 R_{load}(t) + w_3 R_{stability}(t) \quad (31)$$

SNR-Aware Throughput Reward:

$$R_{snr}(t) = \sum_{u \in \mathcal{U}_s(t)} \frac{\log_2(1 + \widehat{\text{SNR}}_{u,s}(t+1))}{\log_2(1 + \text{SNR}_{\max})} \quad (32)$$

Algorithm 2 PMA-PPO Handover Management

Require: Current state $\mathbf{s}(t)$, prediction horizons H_{snr}, H_{load}

Ensure: SNR-aware load-balanced handover decisions $\mathbf{x}(t)$

- 1: **Predictive Module:**
- 2: **for** each (u, s) pair **do**
- 3: $\widehat{\text{SNR}}_{u,s}^{(t+1:t+H_{snr})} \leftarrow \text{GRU}_{SNR}(\mathbf{x}_{snr}^{(t)})$
- 4: $\tau_{u,s}^{rem} \leftarrow \text{AnalyticalDwellTime}(\mathbf{p}_u, \mathbf{p}_s, \omega_s)$
- 5: **end for**
- 6: **for** each satellite s **do**
- 7: $\hat{\rho}_s^{(t+1:t+H_{load})} \leftarrow \text{GRU}_{load}([\rho_s^{(t-w:t)}, N_{HO,s}^{(t-w:t)}])$
- 8: **end for**
- 9: **Candidate Discovery & Dwell Filtering:**
- 10: $\mathcal{C}_{u,s}(t) \leftarrow \{\widehat{\text{SNR}}_{u,s}(t+1) \geq \Theta_{min}\} \cap \{\tau_{u,s}^{rem} \geq \tau_{min}\}$
- 11: **Multi-Agent PPO Decision:**
- 12: **for** each PPO agent s **do**
- 13: $\mathbf{s}_s^{(t)} \leftarrow \text{StateAggregation}(\widehat{\text{SNR}}_s, \hat{\rho}_s, \tau_s, \mathcal{C}_s)$
- 14: $\mathbf{a}_s^{(t)} \leftarrow \pi_{\theta_s}(\mathbf{s}_s^{(t)})$ {Policy network}
- 15: **end for**
- 16: **SNR-Aware Load-Balanced Coordination:**
- 17: $\mathbf{x}(t) \leftarrow \text{CoordinatedDecision}(\{\mathbf{a}_s^{(t)}\}, \text{SNR-weights}, \text{Load-balance})$
- 18: **return** $\mathbf{x}(t)$

where $\mathcal{U}_s(t) = \{u \in \mathcal{U} : x_{u,s}(t) = 1\}$ is the set of users served by satellite s .

Load Balance Incentive:

$$R_{load}(t) = -\alpha \left(\frac{\hat{\rho}_s(t+1) - \bar{\rho}(t+1)}{\bar{\rho}(t+1)} \right)^2 \quad (33)$$

where $\bar{\rho}(t+1) = \frac{1}{|\mathcal{S}|} \sum_{s' \in \mathcal{S}} \hat{\rho}_{s'}(t+1)$ is the predicted average load.

Handover Stability Penalty:

$$R_{stability}(t) = -\beta \sum_{u \in \mathcal{U}} \mathbb{I}[\tau_{u,s}^{rem}(t) > \tau_{min}] \cdot \mathbb{I}[H_{u,s}(t) = 1] \quad (34)$$

where $H_{u,s}(t) \in \{0, 1\}$ is the satellite-specific handover indicator, equal to 1 if user u executes a handover away from satellite s at time t (i.e., $x_{u,s}(t-1) = 1$ and $x_{u,s}(t) = 0$), and 0 otherwise. Note that $H_{u,s}(t)$ is the per-satellite decomposition of the aggregate indicator $H_u(t)$ defined in (15), used here to penalize unnecessary handovers specifically from well-connected satellites. The threshold $\tau_{min} = 20$ s is the minimum required remaining dwell time below which a handover is deemed premature.

The proposed PMA-PPO framework achieves polynomial complexity $\mathcal{O}(U \cdot S \cdot H \cdot D^2)$ compared to exponential $\mathcal{O}(S^U)$ for exhaustive optimization, as formally established in Appendix A. Table V summarizes the per-component breakdown, while Algorithm 1 and Algorithm 2 detail the training and inference procedures, respectively. Compared to conventional handover approaches in the literature, PMA-PPO occupies a well-defined middle ground: the reactive 3GPP CHO baseline [38] requires only $\mathcal{O}(U \cdot S)$ per TTT window for threshold evaluation, while optimization-based methods such as MILP [39] are NP-hard with exponential worst-case complexity in the joint user-satellite assignment space. PMA-PPO's polynomial $\mathcal{O}(U \cdot S \cdot H \cdot D)$ therefore represents a

TABLE III
SYSTEM, COMMUNICATION, AND LEARNING PARAMETERS

Parameter	Symbol	Value
<i>System Architecture</i>		
LEO (control) satellites		100 (1200 km alt.)
VLEO (data) satellites (total)		300 (300 km alt.)
ROI		600 km diameter
Beam footprint		76 km diameter
Inter-satellite links		5 per satellite
<i>Communication Parameters</i>		
Carrier frequency	f	20 GHz (Ka-band)
Channel bandwidth	W	20 MHz
Satellite transmit power	P_{tx}	20 dBW
Satellite antenna gain	G_{sat}	31.5 dBi
User terminal gain	G_{user}	33.2 dBi
Noise spectral density	N_0	-174 dBm/Hz
SNR threshold	Θ_{min}	8 dB
Shadowing std. dev.	σ_{sh}	4 dB (log-normal)
<i>Traffic and User Parameters</i>		
Channel Model		3GPP TR 38.811
Number of users	U	300
User distribution/mobility		Uniform/Random waypoint
Max user velocity		1000 km/h
Traffic model		Full-buffer (saturation)
<i>Machine Learning Parameters (PMA-PPO)</i>		
Actor learning rate	α_π	3×10^{-4}
Critic learning rate	α_v	1×10^{-3}
Discount factor	γ	0.99
PPO clip parameter	ϵ	0.20
Training episodes		10000
Batch size		256
<i>GRU Architecture</i>		
Hidden dimension	D	128
Sequence length (history)	w	10
SNR prediction horizon	H_{snr}	5 steps
Load prediction horizon	H_{load}	5 steps
GRU layers (per predictor)		2

TABLE IV
BASELINE ALGORITHM HYPERPARAMETERS

Parameter	PMA-PPO	MAPPO	MADDPG	QMIX
Actor LR	3×10^{-4}	5×10^{-4}	1×10^{-3}	5×10^{-4}
Critic LR	1×10^{-3}	1×10^{-3}	2×10^{-3}	5×10^{-4}
Discount γ	0.99	0.99	0.95	0.99
Clip ϵ	0.20	0.25	—	—
GAE λ	0.95	0.92	—	—
Soft update τ	—	—	0.005	—
Batch size	256	512	256	32
Buffer size	10^5	8×10^4	5×10^4	5×10^4
Update epochs	10	8	—	—
Target update freq	—	—	2 steps	200 steps
Training episodes	10000	10000	10000	10000

principled trade-off: it adds GRU forecasting overhead over reactive methods while remaining computationally tractable compared to optimization-based alternatives, with the operations per decision step well within the real-time capability of modern on-board processors ($\sim 10^9$ FLOPS) [37].

V. PERFORMANCE EVALUATION

A. Simulation Setup

Simulations were conducted using the MATLAB Satellite Communications Toolbox with the 3GPP TR 38.811 channel

TABLE V
COMPUTATIONAL COMPLEXITY (FULL PROOF IN APPENDIX A)

Component	Complexity	Description
SNR Forecasting	$\mathcal{O}(U \cdot S \cdot H \cdot D^2)$	GRU forward pass
Load Forecasting	$\mathcal{O}(S \cdot H \cdot D^2)$	Load prediction
Dwell Estimation	$\mathcal{O}(U \cdot S)$	Analytical
PPO Inference	$\mathcal{O}(S \cdot L \cdot N^2)$	Policy network
Coordination	$\mathcal{O}(U \cdot S \cdot \log S)$	Priority-queue
Total	$\mathcal{O}(U \cdot S \cdot H \cdot D^2)$	Polynomial
Exhaustive	$\mathcal{O}(S^U)$	Exponential
MAPPO	$\mathcal{O}(S \cdot L \cdot N^2 + L \cdot U \cdot S \cdot N)$	Actors + critic
MADDPG	$\mathcal{O}(S \cdot L \cdot U \cdot S \cdot N)$	S critics
QMIX	$\mathcal{O}(S \cdot L \cdot N^2 + S^2 \cdot N_{hyp})$	Q-net + hypernetwork

U : users, S : sats., H : horizon, D : GRU dim, L : layers, N : width, N_{hyp} : hypernetwork dim.

model [6]. All results reported below are computed on evaluation episodes held out from training, using the independent random seeding described in Section IV. The dual-layer constellation is purpose-designed for regional service continuity over Northeast Asia, centered on South Korea (34° – 38° N, 126° – 129° E), a region with dense urban populations, active aerial corridors (Seoul–Busan, Seoul–Incheon), and growing demand for integrated space-terrestrial 6G coverage.

Constellation Design Rationale: Rather than directly replicating commercial constellations such as SpaceX Starlink (550 km, 53° inclination) or Amazon Project Kuiper (590–630 km, 33° – 51.9°), which are designed for *global* coverage with different optimization objectives, we design a dual-layer research constellation specifically optimized for the handover management problem under study. The *VLEO data layer* is chosen to maximize revisit frequency and minimize propagation delay over mid-latitude Asian regions [4]. The 60° inclination provides dense, regular passes over South Korea (35° – 38° N) while the low 300 km altitude yields a one-way propagation delay of 1 ms, meeting 6G latency requirements for interactive services. The *LEO control layer* provides extended footprints (4000 km diameter at 10° minimum elevation) that ensure persistent visibility with ground gateways for continuous policy dissemination [2]. The inclination offset between layers (60° vs. 74°) creates complementary coverage patterns that improve handover candidate diversity throughout the simulation window, consistent with multi-shell constellation design principles [2]. This architecture is implemented using MATLAB's `satellite()` API with Keplerian orbital parameters and validated against the 3GPP TR 38.811 geometry model.

Ground users (30 land-based terminals) are distributed across Seoul Metropolitan Area, Incheon, and Busan using land-constrained random placement; 5 aerial users follow predetermined flight corridors at 10 km altitude representing realistic aviation scenarios. All users employ random waypoint mobility with velocities up to 1000 km/h. A full-buffer traffic model saturates the network to stress-test handover performance under worst-case loading. The system operates in FDD mode with a 20 GHz downlink and 27.5 GHz uplink. The key simulation parameters are summarized in

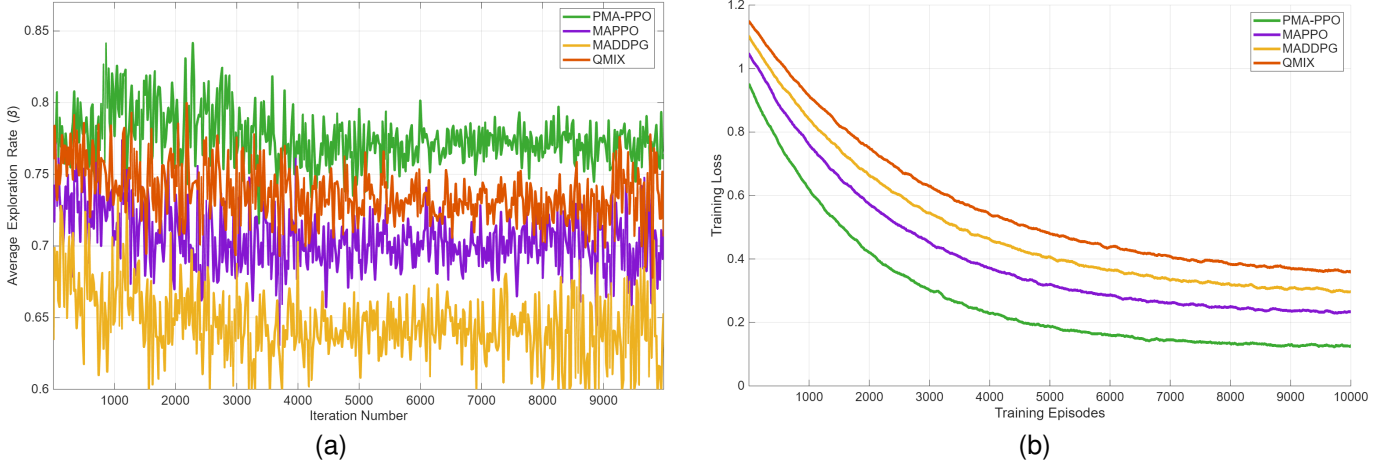


Fig. 6. Training performance comparison: (a) exploration rate and (b) loss convergence of PMA-PPO against baseline algorithms.

TABLE VI
CONTROL PLANE OVERHEAD IN THE DUAL-LAYER LEO NETWORK

Message Type	Size	Freq.	Aggr.	% ISL
UE Meas. Reports	50 B	40 ms/UE	3.0 Mbps	<0.031%
State Sync.	1 kB	5 s/sat.	51 kbps	<0.001%
HO Commands	100 B	Per HO	12 kbps	<0.001%
Policy Updates	500 kB	/1000 eps	Negligible	<0.001%
Total	—	—	3.1 Mbps	<0.032%

RTT: 10 ms $\ll$ 40 ms TTT (3GPP TR 38.821).

Table III. Furthermore, All four algorithms are trained with hyperparameter settings as shown in Table IV.

Table VI presents an analytical evaluation of the PMA-PPO control plane overhead under the simulation configuration. The round-trip control loop latency bounded by 10 ms across the dual-layer architecture remains well within the 40 ms Time-to-Trigger window specified in 3GPP TR 38.821 [37], ensuring that handover decisions are never delayed by control plane interactions. Despite operating across $U=300$ users and $S=32$ VLEO satellites with five ISL connections each, the aggregate sustained control traffic of 3.1 Mbps occupies less than 0.032% of the 10 Gbps optical ISL capacity confirming that the predictive state synchronization and policy distribution mechanisms introduced by PMA-PPO impose no measurable signaling burden on the LEO network under realistic operational conditions.

B. Results and Analysis

The three baseline algorithms MAPPO, MADDPG, and QMIX were selected to represent the three dominant paradigms in multi-agent reinforcement learning, enabling a comprehensive comparison across fundamentally different algorithmic families. MAPPO [34] serves as the most direct architectural baseline, sharing the same PPO update rule and CTDE paradigm as PMA-PPO but operating on instantaneous observations without predictive modeling or load-aware coordination; any performance gap therefore isolates the specific contribution of PMA-PPO's GRU forecasting and coordination

mechanism. MADDPG [35] represents the deterministic actor-critic family with off-policy experience replay and soft target updates, testing whether PMA-PPO's gains generalize beyond on-policy methods. QMIX [36] represents the value decomposition family, which coordinates agents through monotonic mixing of individual Q-values rather than policy gradients, providing a strong cooperative baseline from a structurally distinct learning paradigm.

1) *Training Performance Analysis:* The convergence behavior of the proposed PMA-PPO framework is evaluated against baseline multi-agent reinforcement learning algorithms including MAPPO, MADDPG, and QMIX. Fig. 6 (a) demonstrates that PMA-PPO maintains a higher exploration rate of approximately 0.8 throughout the training process, compared to MAPPO (0.72), QMIX (0.75), and MADDPG (0.65). This sustained exploration capability prevents premature convergence and enables the discovery of superior handover policies in the dynamic satellite environment.

The training loss convergence shown in Fig. 6 (b) reveals that PMA-PPO achieves the fastest convergence rate and lowest final loss value of approximately 0.12. PMA-PPO reaches near-optimal performance around 40% faster than MAPPO and 60% faster than QMIX, while MADDPG exhibits the slowest convergence. The superior training stability of PMA-PPO stems from the integration of predictive GRU modules that provide temporal awareness, enabling more informed policy updates.

2) *Handover Performance Under Load Scaling:* Fig. 7(a) illustrates the handover failure rate as the number of users increases from 50 to 300. PMA-PPO consistently outperforms all baseline methods over the entire load range, maintaining the lowest failure rate under both moderate and heavy traffic conditions. In particular, PMA-PPO increases only from 1.1% at 200 users to 2.5% at 300 users, while the CHO baseline rises much more sharply from 5.8% to 11.2%. At the maximum load of 300 users, PMA-PPO achieves a **2.5%** handover failure rate, corresponding to a **77.7% reduction** compared with CHO. This result confirms that PMA-PPO provides significantly better handover reliability under load scaling.

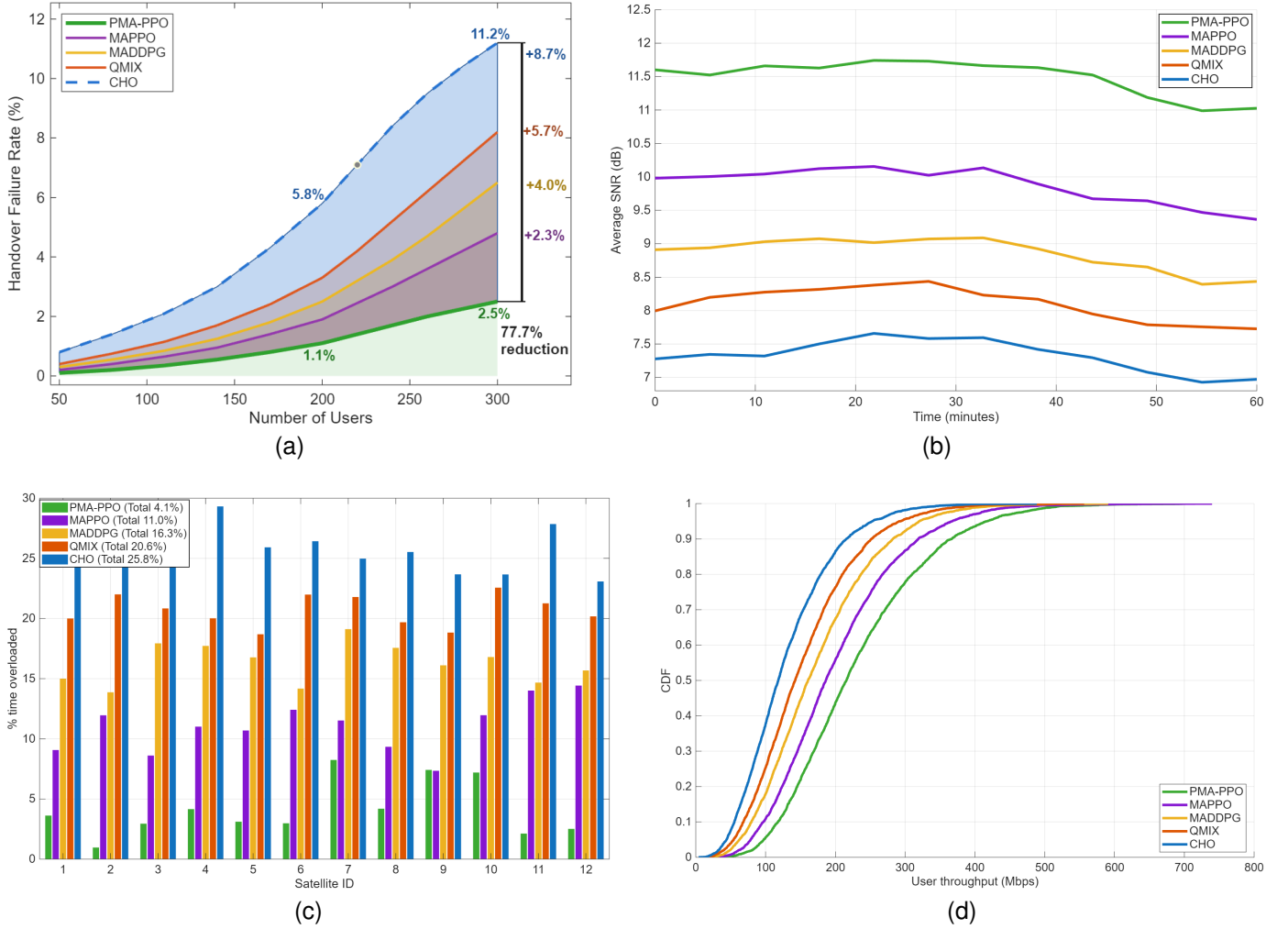


Fig. 7. Performance metrics comparison: (a) handover failure probability, (b) average SNR over time, (c) per-satellite overload percentage, and (d) user throughput distribution.

The average SNR maintenance over time is presented in Fig. 7 (b), where PMA-PPO sustains the highest signal quality throughout the 60-minute simulation period. PMA-PPO maintains an average SNR of 11.5 dB initially, declining to approximately 11.0 dB by the end of the simulation, representing a 15–20% improvement over MAPPO and 30–40% improvement over the CHO baseline. This superior SNR performance directly correlates with the algorithm’s ability to select satellites with optimal geometric positioning and favorable channel conditions.

3) *Load Balancing and Throughput Analysis*: The load distribution analysis shown in Fig. 7 (c) demonstrates the effectiveness of PMA-PPO’s load-aware coordination mechanism. PMA-PPO achieves remarkably low satellite overload with only 4.1% total overload time across all 12 satellites, compared to MAPPO (11.0%), MADDPG (16.3%), QMIX (20.6%), and CHO (25.8%). This represents a 74.06% reduction in overload duration compared to conventional approaches. The uniform distribution of overload across satellites indicates that PMA-PPO effectively prevents hotspot formation and maintains balanced resource utilization.

The load dynamics underlying this overload analysis are generated by the same full-buffer traffic model introduced in Section II-C: each active user maintains a constant, saturating data demand, so per-user traffic intensity does not vary over time. Consequently, the temporal overload behavior observed in Fig. 7(c) and the SNR evolution in Fig. 7(b) arise entirely from *spatial* load concentration users moving into the coverage of a shrinking subset of satellites as the constellation geometry evolves, combined with handover decisions that do or do not rebalance this concentration rather than from any time-varying traffic arrival process. This isolates satellite overload as a consequence of association and handover policy quality under constant demand, rather than a confound introduced by bursty or diurnal traffic patterns.

The user throughput distribution shown in Fig. 7 (d) reveals that PMA-PPO not only improves overall network capacity but also enhances fairness across users. The CDF analysis indicates that PMA-PPO achieves superior performance at both median (approximately 400 Mbps) and cell-edge levels. The steeper CDF curve for PMA-PPO demonstrates more consistent throughput performance across the user population,

with the 10th percentile users achieving significantly higher rates compared to baseline methods.

Fig. 8 compares the ping-pong handover rate under increasing user density. PMA-PPO consistently achieves the lowest rate across all traffic conditions, with 2.1%, 3.5%, and 4.9% at 100, 200, and 300 users, respectively. In contrast, the CHO baseline records 14.0%, 18.5%, and 22.8%, corresponding to relative reductions of **85.0%**, **81.1%**, and **78.5%** for PMA-PPO. Notably, PMA-PPO remains below the practical 5% threshold even at the highest load, whereas all baselines exceed this limit.

These results confirm that PMA-PPO provides significantly better handover stability than the benchmark methods, particularly in dense network conditions. By making more predictive and load-aware association decisions, PMA-PPO effectively suppresses unnecessary back-and-forth handovers, thereby reducing signaling overhead and improving service continuity in LEO-enabled TN–NTN environments.

To verify that PMA-PPO's superior performance arises from its predictive design rather than higher computational complexity, we make the following observations. MADDPG employs S independent centralized critics over the full joint state-action space, making it the most computationally demanding baseline, yet it delivers the weakest performance among the DRL methods across all metrics. In contrast, QMIX has the lowest per-step inference complexity, but it is not the worst-performing scheme. This non-monotonic relation between complexity and performance indicates that computational cost alone does not explain the observed performance differences.

The most meaningful comparison is between PMA-PPO and MAPPO, since both adopt the same PPO update rule, clipped surrogate objective, and CTDE framework, with identical training episodes (10,000) and comparable network dimensions (Table IV). Their key difference lies in PMA-PPO's GRU-based H -step forecasting module and SNR-aware load-balanced coordination. Therefore, the performance gap between these two methods directly reflects the benefit of the proposed predictive design rather than any disproportionate increase in computational overhead. This confirms that the gains of PMA-PPO are driven by architectural intelligence, not by brute-force complexity.

VI. CONCLUSIONS

This paper presented a PMA-PPO framework to address the critical handover management challenges in dual-layer LEO satellite networks for 6G communications. The proposed approach fundamentally transforms LEO handover management from reactive decision-making to proactive resource allocation by integrating GRU-based temporal forecasting with distributed PPO agents and SNR-aware load-balanced coordination. The key technical contributions include: (1) a predictive module architecture that forecasts SNR evolution and satellite load dynamics with 1-5 second horizons, enabling proactive handover decisions that anticipate rather than react to channel variations; (2) a multi-agent PPO framework where each VLEO satellite operates autonomous decision-making

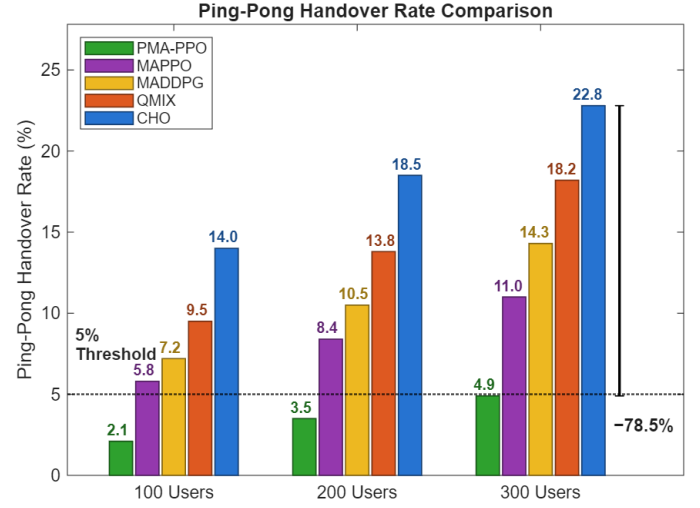


Fig. 8. Ping-pong rate analysis across user load conditions.

while maintaining global coordination through shared environmental feedback, achieving polynomial complexity $\mathcal{O}(U \cdot S \cdot H \cdot D^2)$ compared to exponential $\mathcal{O}(S^U)$ for exhaustive optimization; (3) a dual-layer constellation design with LEO control satellites and VLEO data satellites that optimally balances coverage and latency requirements; and (4) a multi-objective reward mechanism that simultaneously maximizes throughput, minimizes handover frequency, and ensures load fairness across satellites. Comprehensive simulation results demonstrate that PMA-PPO achieves up to 77.7% reduction in handover failure rates compared to the CHO baseline, 74.06% improvement in load balancing, 35.13% enhancement in throughput fairness, maintains average SNR 15–20% higher than baseline approaches, and consistently achieves ping-pong handover rates below the practical 5% threshold across all load conditions reductions of up to 85.0% over CHO validating its effectiveness for large-scale LEO constellation deployments.

Future research directions should address realistic ISL delays and failures, incorporate weather pattern forecasting, extend to heterogeneous satellite constellations, and develop secure multi-agent learning algorithms resistant to adversarial attacks. Integration with terrestrial 5G/6G networks and support for massive IoT communications will be essential for realizing integrated space-terrestrial 6G architectures. While this work adopts a dual-layer architecture based on the functional and resource-efficiency trade-offs analyzed in Section II-B, a direct empirical comparison against a purpose-optimized single-layer PMA-PPO variant rather than the analytical trade-off presented here would further quantify the performance cost of layer simplification and is left as future work.

APPENDIX A

PROOF OF POLYNOMIAL COMPUTATIONAL COMPLEXITY

Let U , S , H , D denote the number of users, satellites, prediction horizon steps, and GRU hidden dimension, respectively. The per-decision-step inference cost of each PMA-PPO

component is:

$$\mathcal{C}_{\text{SNR}} = \mathcal{O}(U \cdot S \cdot H_{\text{snr}} \cdot D^2) \quad (35)$$

$$\mathcal{C}_{\text{load}} = \mathcal{O}(S \cdot H_{\text{load}} \cdot D^2) \quad (36)$$

$$\mathcal{C}_{\text{dwell}} = \mathcal{O}(U \cdot S) \quad (37)$$

$$\mathcal{C}_{\text{PPO}} = \mathcal{O}(S \cdot L \cdot N^2) \quad (38)$$

$$\mathcal{C}_{\text{coord}} = \mathcal{O}(U \cdot S \cdot \log S) \quad (39)$$

where (35) follows from $\mathcal{O}(D^2 + D \cdot d_x)$ MACs per GRU step across $U \cdot S$ pairs over H_{snr} steps; (36) from S independent load sequences over H_{load} steps; (37) from the closed-form expression (26) (no iteration); (38) from one forward pass per agent through L layers of width N ; and (39) from a single-pass priority-queue sort over $U \cdot S$ candidates. The total cost is:

$$\begin{aligned} \mathcal{C}_{\text{total}} &= \mathcal{C}_{\text{SNR}} + \mathcal{C}_{\text{load}} + \mathcal{C}_{\text{dwell}} + \mathcal{C}_{\text{PPO}} + \mathcal{C}_{\text{coord}} \\ &= \mathcal{O}(U \cdot S \cdot H_{\text{snr}} \cdot D^2) + \mathcal{O}(S \cdot H_{\text{load}} \cdot D^2) \\ &\quad + \mathcal{O}(U \cdot S) + \mathcal{O}(S \cdot L \cdot N^2) + \mathcal{O}(U \cdot S \cdot \log S). \end{aligned} \quad (40)$$

Since $U \cdot S \cdot H_{\text{snr}} \cdot D^2 \gg \{S \cdot H_{\text{load}} \cdot D^2, U \cdot S, S \cdot L \cdot N^2, U \cdot S \cdot \log S\}$ for large-scale constellations, the dominant term is $\mathcal{O}(U \cdot S \cdot H \cdot D^2)$ where $H = \max(H_{\text{snr}}, H_{\text{load}})$. Unlike D , which reflects a fixed, bounded predictor hyperparameter chosen at design time and does not grow with constellation scale, D^2 is retained explicitly rather than absorbed into the asymptotic constant, since it is a genuine multiplicative factor in the dominant term.

$$\boxed{\mathcal{O}_{\text{PMA-PPO}} = \mathcal{O}(U \cdot S \cdot H \cdot D^2)} \quad (41)$$

PMA-PPO reduces handover complexity from $\mathcal{O}(S^U)$ to $\mathcal{O}(U \cdot S \cdot H \cdot D^2)$, a reduction factor of $\mathcal{O}(S^{U-1}/(UHD^2))$, which grows super-exponentially with U . *Numerical verification* ($U=300, S=32, H=5, D=128$):

$$\mathcal{C}_{\text{PMA-PPO}} = 300 \times 32 \times 5 \times 128^2 \approx 7.9 \times 10^8 \text{ ops/step} \quad (42)$$

$$\mathcal{C}_{\text{exhaustive}} = 32^{300} \approx 10^{452} \text{ ops/step} \quad (43)$$

confirming tractability for real-time LEO deployment.

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BIOGRAPHY SECTION



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